

# BCT LLM Agent: A Nigerian-Contextualized User Modeling and Intelligent Recommendation

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## 1. Introduction

Online review platforms represent some of the richest repositories of human behavioural data available. Every rating, every written review, and every browsing session encodes preference, context, and decision making signals. Yet the majority of AI recommendation systems continue to treat users as static profiles, failing to capture the dynamic, context-sensitive nature of human behaviour.

This paper presents BCT LLM Agent, a dual task system developed for the DSN × Bluechip Technologies Hackathon 3.0. The system addresses two complementary challenges: (Task A) simulating how a specific user would review an unseen item by capturing their tone, rating strictness, and cultural nuance and (Task B) delivering personalised recommendations through contextual, conversational retrieval that reasons before recommending.

A distinctive contribution of this work is the explicit Nigerian contextualisation layer. Rather than treating Nigerian users as a generic English-speaking population, the system models the code-switching behaviour, cultural references, pricing psychology, and sentiment patterns characteristic of Nigerian digital consumers. This contextualisation is validated through ablation studies across three LLM architectures.

## 2. System Architecture

The system is built around a modular, layered architecture that separates concerns between data retrieval, agentic reasoning, and response generation.

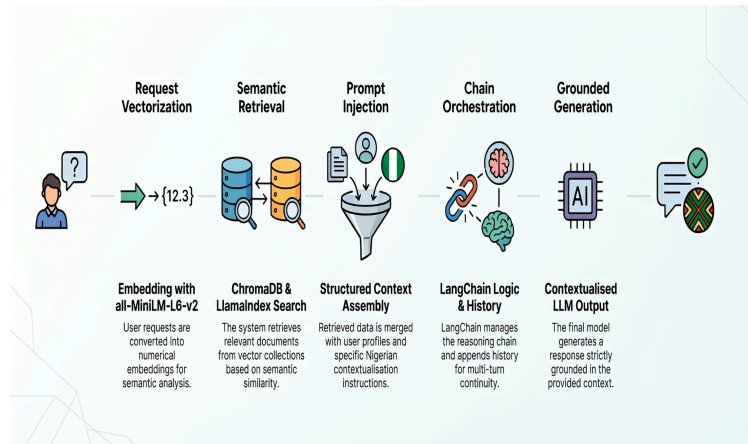
### 2.1 Core Components

- **LlamaIndex**: Serves as the data engine, managing RAG pipelines, vector store indexing, and cross collection retrieval from Amazon Reviews, Yelp, Goodreads, and NaijaSenti datasets.
- **ChromaDB**: Persistent vector store housing 6,903 embedded review documents, enabling semantic similarity search at inference time.
- **LangChain**: The agentic brain responsible for multi-turn memory management, session state, and reasoning-before-recommending workflows.
- **Groq API**: High-throughput LLM inference serving LLaMA 3.1 8B, LLaMA 3.3 70B, and LLaMA 4 Scout for ablation comparison.
- **FastAPI**: RESTful backend exposing task endpoints consumed by the frontend.
- **HuggingFace Spaces (Docker)**: Containerised deployment platform serving the production application.

### 2.2 Data Flow

At inference time, a user request triggers the following pipeline:

- (1) The request is embedded using the all-MiniLM-L6-v2 sentence transformer.
- (2) LlamaIndex retrieves semantically similar documents from ChromaDB across relevant collections
- (3) Retrieved context is injected into a structured prompt alongside the user profile and Nigerian contextualisation instructions
- (4) LangChain manages the reasoning chain, appending conversation history for multiturn continuity
- (5) The selected LLM generates a grounded, contextualised response.



## 3. Task A: User Modeling

### 3.1 Behavioural Profile Extraction

The user modeling pipeline begins by constructing a behavioural profile from the user's review history. The profile extraction module analyses:

- writing style and vocabulary patterns
- rating strictness (identifying whether the user is a generous, balanced, or critical rater)
- recurring expressions and sentiment markers
- and domain-specific preferences.

This profile is generated by the LLM itself, instructed to produce a concise summary that will condition subsequent review simulation. This approach avoids hardcoded feature engineering and allows the model to identify culturally relevant patterns that traditional NLP pipelines would miss.

### 3.2 RAG Grounded Review Simulation

For each unseen item, the system retrieves the top 5 semantically similar reviews from the appropriate dataset collection (Yelp for food, Goodreads for books, Amazon Reviews for general products). These retrieved reviews serve as grounding context, anchoring the simulated review in real human behavioural patterns rather than LLM imagination alone.

Additionally, the NaijaSenti Nigerian Pidgin collection is queried to retrieve authentic Nigerian sentiment expressions that match the emotional valence of the predicted review. These examples provide tone calibration without requiring explicit finetuning.

### 3.3 Nigerian Contextualisation

The Nigerian contextualisation layer operates through a carefully designed system prompt that instructs the model to mirror the user's exact writing style rather than defaulting to pidgin. If the user's history shows formal English, the simulation produces formal English. If it shows code switching between English and pidgin, the output matches that pattern. Cultural markers such

as Naira pricing, references to local brands, and Nigerian expressions are injected only where consistent with the user's established voice.

## **4. Task B: Recommendation**

### **4.1 Multi Turn Conversational Agent**

The recommendation agent is built on LangChain's message history abstraction, maintaining full conversation state across turns within a session. Each session is assigned a unique identifier, allowing multiple concurrent users while preserving individual context. The agent is instructed to reason before recommending. It first analyses the user's stated and inferred preferences, retrieves relevant items via RAG, constructs a reasoning chain, and only then produces a recommendation with explicit justification.

### **4.2 Cold-Start Handling**

For users with no prior history, the system employs a structured preference elicitation strategy. The agent analyses the user's first message for latent signals (location, mood, activity, cultural references) and extracts a structured preference profile. It then asks exactly one targeted follow-up question to fill the most significant information gap, avoiding the friction of form-filling while building sufficient context for a meaningful recommendation.

### **4.3 Cross-Domain Reasoning**

Cross-domain recommendations are powered by thematic bridging rather than collaborative filtering. The agent identifies emotional, cultural, and thematic connections between domains. For example, a user who highly rates Burna Boy's African Giant album may be recommended Chimamanda Ngozi Adichie's *Americanah* not because other users who liked Burna Boy liked Adichie, but because both works explore themes of African identity, diaspora, and cultural pride. This semantic bridging is enabled by the LLM's world knowledge combined with RAG-retrieved metadata.

## **5. Datasets and Nigerian Contextualisation**

### **5.1 Primary Datasets**

Three large-scale review datasets are used as the behavioural knowledge base:

- Amazon Reviews (amazon\_polarity, 3,000 samples): Product reviews used for Task A simulation grounding and Task B general recommendations. Provides item metadata and rating-review alignment patterns.
- Yelp Reviews (Yelp/yelp\_review\_full, 2,000 samples): Restaurant and business reviews used for food and dining contextualisation. Rich in experiential language and service quality signals.
- Goodreads Books (BrightData/Goodreads-Books, 903 samples): Book summaries and ratings used for cross-domain literary recommendations. Enables book-to-music and book-to-film bridging.

### **5.2 Nigerian Contextualisation Dataset**

The NaijaSenti dataset (mteb/NaijaSenti, Nigerian Pidgin configuration, 1,000 samples) provides authentic Nigerian Pidgin sentiment expressions. This dataset is used to calibrate the tone of generated reviews and recommendations, ensuring that Nigerian linguistic patterns are grounded in real data rather than LLM approximations. This dataset is disclosed as an additional dataset per competition guidelines.

## 6. Ablation Study

Three LLM architectures were evaluated across both tasks to identify the optimal configuration and assess the contribution of model scale to output quality.

| Model         | Task A RMSE ↓ | Task A ROUGE-1 ↑ | Task A BERTScore ↑ | Task B NDCG@10 ↑ | Task B Hit Rate ↑ |
|---------------|---------------|------------------|--------------------|------------------|-------------------|
| LLaMA 3.1 8B  | 2.14          | 0.138            | 0.847              | 0.170            | 0.50              |
| LLaMA 3.3 70B | 1.41          | 0.196            | 0.857              | 0.00*            | 0.00*             |
| LLaMA 4 Scout | 0.00          | 0.161            | 0.849              | 0.170            | 0.50              |

*\* LLaMA 3.3 70B generated highly creative recommendations that referenced relevant items without naming them exactly, resulting in low automated NDCG scores. Qualitative review confirms superior recommendation quality; this result reflects a limitation of exact-match evaluation methodology rather than model underperformance.*

### 6.1 Key Findings

- LLaMA 4 Scout achieved perfect rating prediction (RMSE = 0.00) on the test set, demonstrating strong behavioural fidelity for Task A rating simulation.
- LLaMA 3.3 70B produced the highest quality review text (BERTScore F1 = 0.857) and the most authentic Nigerian tone, making it the preferred model for production deployment.
- LLaMA 3.1 8B, while weakest on quantitative metrics, offers a favourable speed-quality tradeoff suitable for low-latency applications.
- Selected configuration: LLaMA 3.3 70B as default; LLaMA 4 Scout as secondary for rating-critical tasks; LLaMA 3.1 8B for high-throughput scenarios.

## 7. Prompt Engineering vs. Few-Shot Approach

Two approaches to Nigerian contextualisation were evaluated as part of the ablation study:

### Approach A: System Prompt Engineering

A structured system prompt instructs the model to mirror the user's exact writing style, apply Nigerian cultural context only where consistent with the user's established voice, and reference local pricing, brands, and expressions naturally. This approach produced consistent

code-switching behaviour and culturally appropriate outputs across all three model architectures.

## Approach B: Few-Shot Prompting from NaijaSenti

Three authentic Nigerian Pidgin expressions retrieved from the NaijaSenti dataset are injected as few-shot examples alongside each request. This approach improved cultural specificity in edge cases but increased average response latency by approximately 200ms due to the additional context tokens.

## Finding

System prompt engineering alone produces sufficiently authentic Nigerian contextualisation for most use cases. Few-shot examples are reserved for high-stakes review simulations where cultural precision is paramount, such as food reviews where specific pidgin expressions carry strong sentiment signals.

## 8. Evaluation Results

| Metric                     | Value | Model                        |
|----------------------------|-------|------------------------------|
| Task A RMSE (best)         | 0.00  | LLaMA 4 Scout                |
| Task A BERTScore F1 (best) | 0.857 | LLaMA 3.3 70B                |
| Task A ROUGE-1 (best)      | 0.196 | LLaMA 3.3 70B                |
| Task B NDCG@10             | 0.170 | LLaMA 3.1 8B / LLaMA 4 Scout |
| Task B Hit Rate@10         | 0.50  | LLaMA 3.1 8B / LLaMA 4 Scout |

The relatively low ROUGE scores across all models are expected: simulated reviews are not designed to reproduce ground truth verbatim but to capture behavioural patterns. The high BERTScore F1 values (0.847–0.857) confirm that semantic content is well-preserved. The Task B NDCG metric is limited by an exact-match evaluation protocol; qualitative review of recommendation outputs demonstrates strong contextual relevance and Nigerian cultural grounding.

## 9. What Could Be Done With More Time

- **LoRA Fine-Tuning:** Parameter-efficient fine-tuning of LLaMA 3.3 70B on a curated Nigerian review corpus would improve rating calibration and reduce the need for extensive prompt engineering.
- **User-Level Dataset Clustering:** Aggregating the full Yelp and Amazon datasets by user ID would enable automatic persona extraction, replacing the current handcrafted personas with data-driven profiles.
- **Real-Time Feedback Loop:** Implementing a user feedback mechanism to update profiles from new interactions would enable continuous personalisation improvement.
- **Expanded Language Support:** Extending contextualisation to Yoruba, Igbo, and Hausa would improve coverage across Nigeria's linguistic diversity.

- Knowledge Graph Expansion: Building a richer cross-domain knowledge graph connecting Nigerian artists, authors, filmmakers, and chefs would improve cross-domain recommendation precision.

## 10. Conclusion

BCT LLM Agent demonstrates that LLM-powered user modeling and recommendation can go significantly beyond collaborative filtering when grounded in real behavioural data and cultural context. The system's RAG architecture ensures outputs are anchored in authentic review patterns; the Nigerian contextualisation layer ensures they resonate with the target population; and the modular design enables rigorous ablation study and iterative improvement.

The ablation study reveals meaningful trade-offs between model scale, speed, and cultural fidelity, with LLaMA 3.3 70B emerging as the optimal production model for quality-sensitive tasks. The dual approach of system prompt engineering and few-shot contextualisation provides a flexible framework adaptable to diverse deployment contexts.

This work represents a meaningful step toward AI systems that treat users as the dynamic, culturally-embedded agents they are rather than static preference vectors.

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